ASSIGNMENT 2

TYBBA(CA) / TYBCA –

Chapter 2: Cloud Deployment Models and Virtualization

SECTION A: Short Notes - 5 Marks Each

(Attempt any 5)

Q1. What is a Cloud Deployment Model? Explain its need. (IMP - Asked 2023)

Q2. Write short note on Public Cloud with features, advantages & disadvantages. *(VERY IMP - 90% chance)*

Q3. Write short note on Private Cloud vs Public Cloud - Differentiate.

Q4. What is Virtualization in Cloud Computing? Explain with example. *(VERY IMP - 2024)*

Q5. What is a Hypervisor? Explain Type 1 [Bare Metal] and Type 2 with examples like VMware ESXi, VirtualBox. *(MOST IMP)*

Q6. Differentiate between Virtual Machines and Containers.

Q7. Write short note on Docker and Kubernetes - Overview and Difference. *(VERY IMP)*

Q8. What is Resource Pooling? Explain Benefits and Limitations.

Q9. What is Multi-Tenancy? Explain with Gmail/Salesforce example.[Hosted][VMs]

SECTION B: Long Answers - 10/15 Marks

(Attempt any 5)

Q1. [MOST REPEATED - 2022,2023,2024] Explain 4 Cloud Deployment Models - Public, Private, Hybrid, Community Cloud with features, advantages, disadvantages and examples [AWS, SBI Bank, Flipkart].

Q2. Explain Virtualization with working. Explain Types: Server, Storage, Network, Desktop, Application, Data, Full, Para, OS-Level Virtualization.

Q3.Explain Hypervisor in detail. Explain Type 1 and Type 2 with architecture diagram, examples [VMware ESXi, Hyper-V, VirtualBox, Workstation].

Q4. Explain VMs vs Containers. Explain Docker with features, components, benefits and Kubernetes architecture, features, components. Differentiate Docker and Kubernetes.

Q5. Explain Resource Pooling and Multi-Tenancy with advantages/disadvantages. Explain better resource utilization, lower cost, security risks, performance issues, single point of failure.

Q6. Explain Hybrid Cloud and Community Cloud with advantages/disadvantages. Why choose Hybrid over Public?